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THE NUMBER OF HOURS TAKEN TO TRANSMIT THE FIRST MESSAGE ON THE 1858 CABLE



Workers inspecting the transatlantic telegraph cable. The first message sent was "Glory to God in the highest; on earth, peace and good will toward men."

1856, quarry workers found some bones in a cave and local teacher **Johan Fuhlrott** (1803–77) identified them as humanlike. Named after the Neander Valley in Germany where the bones were found, this human ancestor is now known as **Neanderthal man** (*Homo neandertalis*). Neanderthal man is thought to have lived in Europe between 300,000 and 30,000 years ago, but it was some time before many would accept that any other humanlike creatures had ever lived.

In America, meteorologist **William Ferrel** (1817–91) explained how **rising warm air** and Earth's rotation creates spiralling circulations of air in the mid-latitudes, called **Ferrel cells**. These cells drive the stormy, westerly winds that are characteristic of these latitudes.

In India, British surveyor **Thomas Montgomerie** (1830–78) began his **survey of the Karakoram Range** as part of the Great Trigonometric Survey of India that had started in 1802.

British inventor Alexander Parkes (1830–90) patented the **first plastic**, Parkesine, which was made from cellulose treated with acid and a solvent.

IN FRANCE, IN 1857, CHEMIST AND MICROBIOLOGIST LOUIS PASTEUR (1822–95) published his seminal results on **fermentation** and yeast multiplication in 1857. He found that when beer and wine ferment it is not chemicals that are responsible, but **tiny microbes** known as yeast. Pasteur later



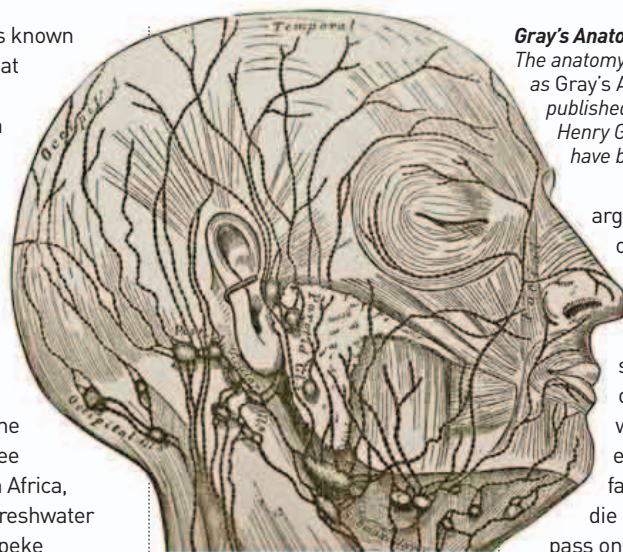
ALFRED WALLACE
(1823–1913)

Alfred Wallace is best known for independently conceiving a theory of evolution by natural selection. He also pioneered biogeography, the study of species in a geographical area. His work in Indonesia between 1854 and 1862 led to the Wallace Line, which divides Asian and Australian species (see pp.204–205).

developed a process known as pasteurization that involved killing the yeast microbes with heat in order to prolong the life of certain foods.

On February 13, 1858, British explorers **Richard Francis Burton** (1821–90) and **John Hanning Speke** (1827–64) became the first Europeans to see **Lake Tanganyika** in Africa, the second largest freshwater lake in the world. Speke continued alone to discover Lake Victoria.

In the cities, **fast-rising populations** were putting new demands on engineers and builders. In New York, **Elisha Otis** (1811–61), inventor of a special safety device that prevented lifts from falling if the cables failed, installed his **first elevator** at 488 Broadway on March 23, 1857. In Germany, Friedrich Hoffmann played his part in speeding up urbanization by patenting the Hoffmann kiln in 1858, which was capable of **firing bricks non-stop**. Cities also began to communicate across oceans. The first undersea **transatlantic telegraph cable**, laid between western Ireland and



Gray's Anatomy

The anatomy book now known as Gray's Anatomy was first published by British surgeon Henry Gray in 1858. There have been over 40 editions.

argued that all species on Earth had evolved gradually due to a process of change known as natural selection—members of the species less well suited to the environment either fail to reproduce or die early, thus failing to pass on their inferior traits.

Wallace had written to Darwin from Indonesia in June 1858 with an outline of his idea, but, unbeknown to him, Darwin had already spent two decades developing the theory.

The most widely used book in medical history was published in 1858. British surgeon Henry Gray's (1827–61) *Anatomy: Descriptive and Surgical* has been in publication ever since and is now known simply as **Gray's Anatomy**.

Newfoundland in Canada, went into service in August 1858.

On July 1, 1858, a scientific paper was delivered to the Linnean Society in London. It combined the ideas of the British naturalists Charles Darwin and Alfred Russel Wallace in a new theory—the theory of **evolution by natural selection**. The idea that species evolved through time was not new, but Darwin and Wallace

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MILES THE LENGTH OF LAKE TANGANYIKA IN AFRICA

British inventor Alexander Parkes patents the first plastic, Parkesine

March 23, 1857
Elisha Otis's first elevator is installed in New York

1857 French chemist Louis Pasteur shows that fermentation is caused by living microbes

May 14, 1858
British missionary David Livingstone's African Zambesi expedition begins

July 1, 1858
British naturalists Darwin and Wallace's theory of evolution by natural selection is presented

August 3, 1858
John Speke reaches Lake Victoria in Africa

June 1858
Wallace writes to Darwin from Indonesia

August 16, 1858
First transatlantic telegraph cable message is sent

1858
First Hoffmann brick kiln, capable of firing bricks non-stop, patented in Germany